Proposal Defense

Socratic AI, Agentic AI for Fostering Critical Thinking Education

A multi-agent tutor that asks instead of answers, and measures how well the learner reasons

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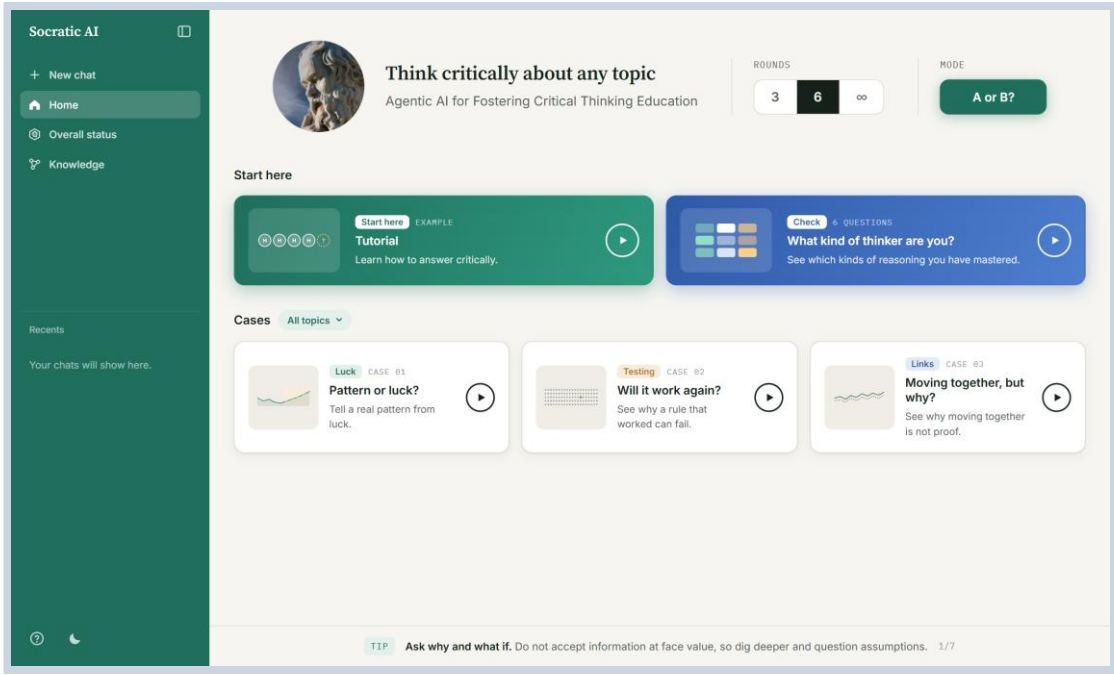
Course EFB4042 Final Year Project I

Project code 26S005

Date 28 October 2026

OUTLINE OF PRESENTATION

1. Introduction
2. Problem Statement, Objectives, Scope of Study
3. Literature Review
4. Proposed Methodology
5. Gantt Chart & Milestones
6. Preliminary Results
7. Conclusion



The working prototype, live online

INTRODUCTION

Students can now ask AI for an answer and copy it, so some skip the thinking.

62% of students

use generative AI at least weekly [12]

319 workers studied

says more trust in AI meant less critical thinking [10]

4 costs of "outsourced reasoning"

- fewer ideas,
- less effort,
- weaker memory,
- less reflection [11]

The idea of this project

Turn the AI around. The tutor asks questions in a set order, so the student does the thinking, and then it measures that thinking.

PROBLEM STATEMENT

AI now does much of the analysis and coding, so engineering students need strong critical thinking [10]. Socratic AI tutors avoid giving answers, but they have three gaps.

1

No ladder

The questions only lead to an answer, never to whether it is the right question.

2

No measurement

They do not measure the learner's critical thinking against any framework.

3

No adaptation

They do not adapt to what each learner is missing. One tester wrote "I've explained this before."

For example, a student asks a chatbot why a circuit keeps overheating, copies the confident fix, and never asks what caused it.

So engineering students need an agentic AI tool that builds their critical thinking.

OBJECTIVES

1. Undergraduates who are new to a body of knowledge reach higher levels of Paul's six levels of Socratic questioning, without the tutor giving them the answer.

2. The tutor's marks of each learner's critical thinking, on Paul and Elder's nine standards and on the type of reasoning and knowledge used, agree with human markers.

3. Each learner gets questions and hints that match their gaps, and can see which skills they have mastered and which still need work.

Each objective can be judged as met or not met at the end of the project.

SCOPE OF STUDY

1

Learners

Beginner undergraduates at UTP.
Short English text chats.

2

Knowledge

Any body of knowledge as text.
A few topics from different fields.
Quant checked by hand.

3

Critical thinking

As defined by Paul and Elder.
Measured only inside the sessions.

Outside the scope are long-term change in how learners think and a full class study.

These belong to the later phases of the parent grant.

LITERATURE REVIEW

Agentic AI Building Blocks

Technique	Year	Core idea	Use in this project	Ref
Prompting	2025	Instruction plus context, and the context shapes the output	Every agent gets a structured JSON prompt	[15]
Chain of Thought	2022	Break a task into intermediate reasoning steps	The Planner reasons before choosing the next move	[16]
ReAct	2023	Reason, act, observe, then repeat	The Tutor runs a simulation, reads the result, then replies	[17]
RAG	2023	Ground answers in retrieved knowledge (naive, advanced, modular)	Case notes now; RAG with a knowledge graph planned for FYP2	[14], [18]
Model Context Protocol	2024	A standard way to give an LLM tools, resources and prompts	Later, to query a knowledge graph and learner records	[19]
Multi-agent systems	2024	Specialised agents collaborate (MetaGPT, AgileCoder)	Assessor, Planner, Tutor and Checker roles	[20], [21]

Based on Dr Ho's course slides [15], [22] and the DeepLearning.AI courses [23].

LITERATURE REVIEW

Frameworks for Critical Thinking

Framework	Year	Core idea	Role in this project	Limitation	Ref
Paul's 6 Socratic question levels	2006	Clarify, assumptions, evidence, viewpoints, implications, question the question	The ladder of what the Tutor asks next	Says what to ask, not how good the answer is	[1]
Paul-Elder 9 intellectual standards	2019	Clarity, accuracy, precision, relevance, depth, breadth, logic, significance, fairness	The Assessor marks each as 1 or 0	Does not say which type of reasoning was used	[1]
Paul-Elder 8 elements of thought	2019	Purpose, question, information, inference, concepts, assumptions, implications, point of view	Prerequisites before reasoning (planned)	Links to reasoning types still open	[1]
Revised Bloom's taxonomy	2001	Knowledge types, such as factual, conceptual, procedural and metacognitive	Types of knowledge used (planned)	No causal or relational type	[2]
SOLO taxonomy	1982	Prestructural to extended abstract outcomes	Complexity, by how many knowledge types are joined	Needs adapting to dialogue	[3]

LITERATURE REVIEW

Socratic Questioning and Socratic AI

Study	Year	Setting	Main finding	Limitation	Ref
Sahamid	2016	Classroom action research	Socratic questioning built critical thinking	Small, one class	[4]
Makhene	2019	Nursing education	Socratic inquiry helps critical thinking	Needs skilled teachers' time	[5]
Ho, Chen and Li	2023	Healthcare students	Socratic method improved thinking skills	Human tutor, does not scale	[6]
Degen and Asanov	2025	65 pre-service teachers, Socratic AI vs plain chatbot	More support for critical, reflective thinking; argues for multi-agent systems	Self-reported, not a direct measure	[7]
Fan	2025	Socratic scaffolding through AI	Structured scaffolding deepened inquiry	No link to reasoning constructs	[8]
Lee, Ang and Lai	2025	Bloom labels on chatbot dialogues	Dialogue logs reveal thinking processes	Descriptive only	[9]
Lee et al. (CHI)	2025	319 knowledge workers	More AI trust, less critical thinking	Survey, self-reported	[10]

LITERATURE REVIEW

Existing Socratic AI Tutors

Tool	Refuses answers	Question ladder	Framework assessment	Goal, progress, end	Reasoning and knowledge types
Online Gemini Gem "Socratic Tutor" (Test #1)	Yes	No	No (informal points)	No	No
Project prompt v1 (Test #2)	Yes	Partly	Partly (hidden 1 to 3 score)	No	No
Khanmigo, custom GPTs *	Yes	Not shown	Not shown	Partly	No
Socratic AI (this project)	**Yes**	**Yes, 6 levels**	**Yes, 9 standards**	**Yes**	**Yes**

** From public descriptions, not tested in this project.*

No tested tool has a full question ladder, framework marks and a clear session, and none shows the type of reasoning.

LITERATURE REVIEW

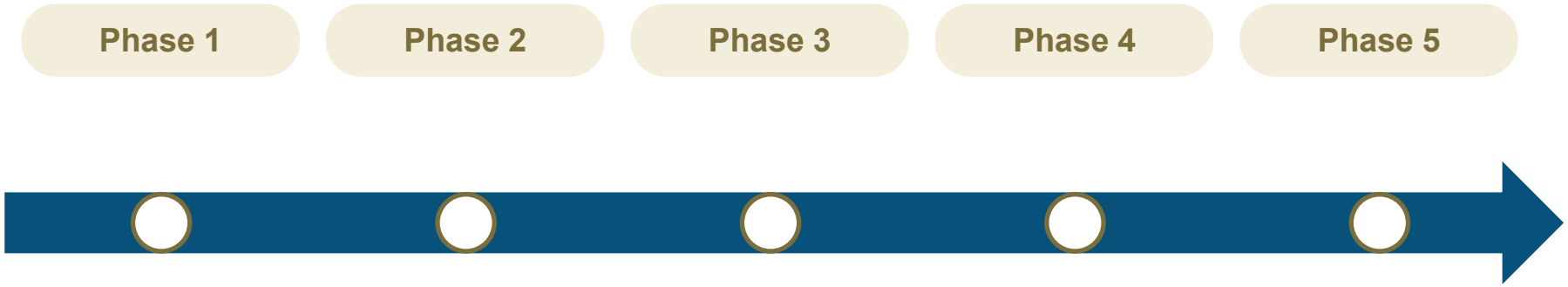
RESEARCH GAP

- AI can make students think less, and most AI tutors just give answers.
- Socratic AI tutors ask questions, but in no clear order.
- Most studies use what students say, not measured reasoning.
- No tool shows the type of reasoning or knowledge used.

JUSTIFICATION

- The Tutor follows Paul's six levels, so questions go up in a set order.
- A separate Assessor marks each answer on nine standards with evidence, so we can compare it with human markers.
- Code rules check every AI decision, so each step is fair and testable.
- A Reasoning agent now marks the type of reasoning and knowledge.

PROPOSED METHODOLOGY



Early testing

Two Gemini Gem tutors tested; problems written down

Requirements

9 design requirements R1 to R9 from the tests

Build

AI agents, rules, cases, simulations, web app

Test and fix

Real sessions read turn by turn; problems fixed

Evaluate

Requirements, Assessor vs humans, learners

R1 Clear mission	R2 Fixed rounds, progress	R3 Remember, never re-ask
R4 Feedback and teaching	R5 Notice frustration	R6 Vary formats
R7 Separate assessor	R8 Learner stats	R9 Stable topic

Software

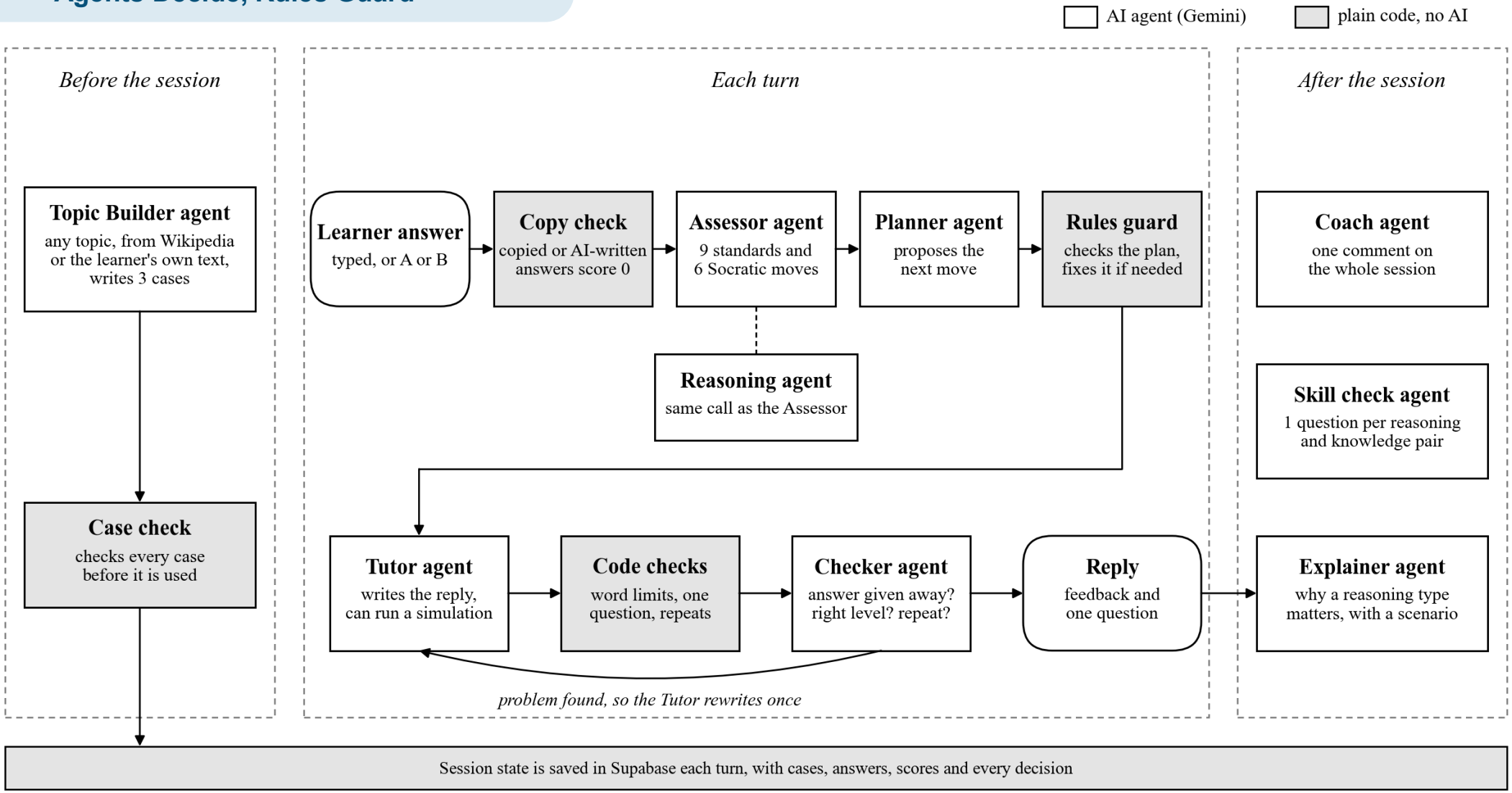
- Python 3.13, FastAPI, LangGraph
- Free Gemini 3.5 Flash Lite now, Claude AI supported
- HTML and JavaScript front end
- Supabase database, Render hosting (free)
- 196 automatic tests with pytest

Hardware

- A normal laptop, no GPU needed

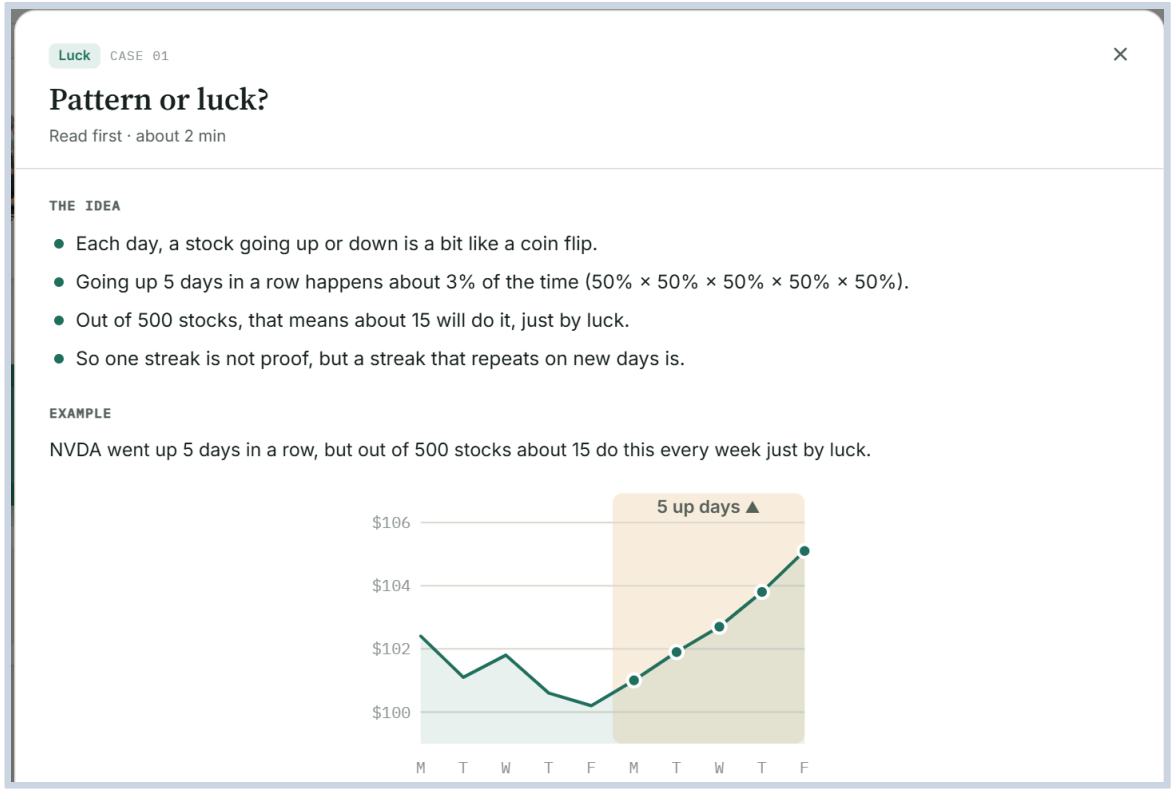
PROPOSED METHODOLOGY

Agents Decide, Rules Guard

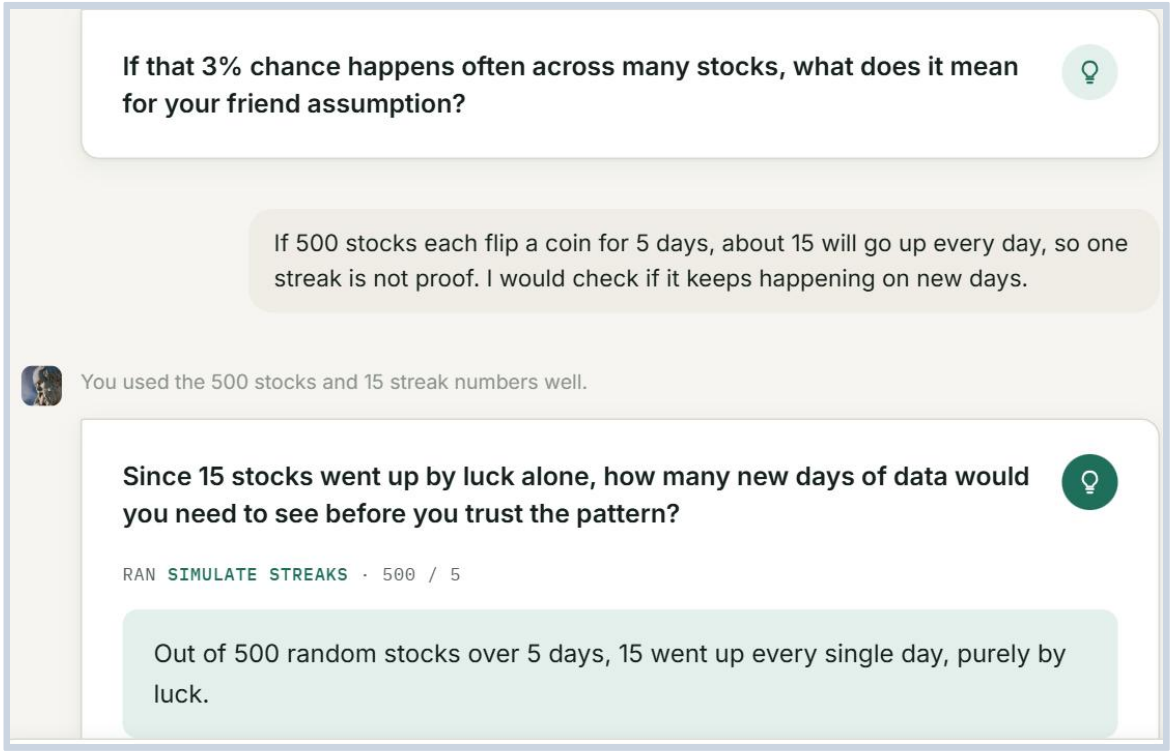


PROPOSED METHODOLOGY

Session Design (R1 to R9)



1. A short reading and a mission before each case

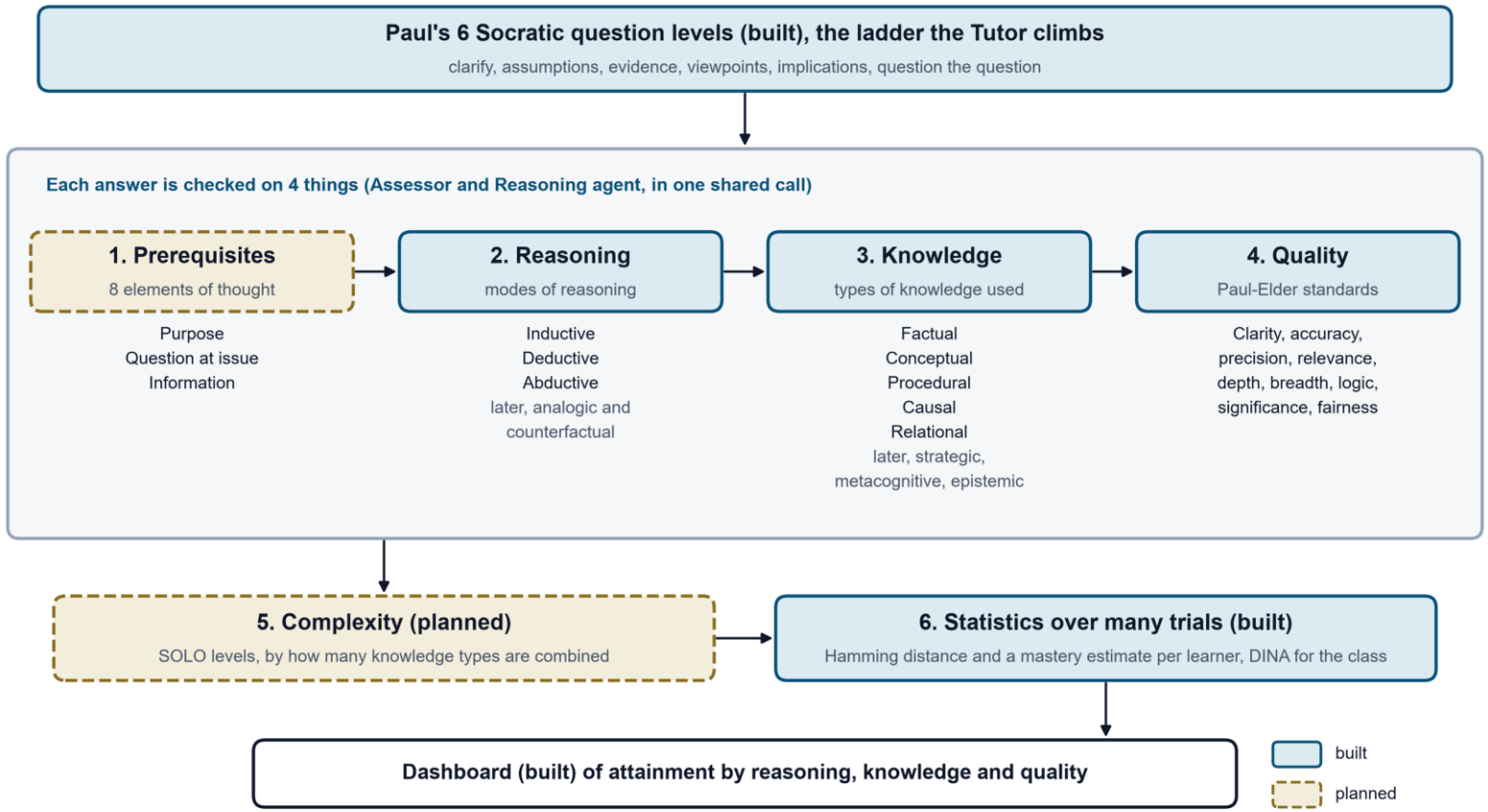


2. Feedback, one question, a hidden hint, and a simulation result

3, 6 or no-limit rounds | **Round and level shown** | **4 question formats** | **"I'm not sure" button** | **Status panel** | **Tutorial**

PROPOSED METHODOLOGY

Reasoning and Knowledge Layer (Goal 1, built)



- **The Tutor climbs Paul's 6 levels.** The student answers each question.
- **1 to 4 check each answer.** Basics, reasoning, knowledge and quality each get a 1 or 0.
- **Complexity (planned)**. One knowledge type is simple, and three joined is complex.
- **Statistics**. Many answers show which skills are mastered, and the Hamming distance [13] counts the skills left.

For example, "All the regional banks fell, not just CIMB" is inductive reasoning on relational knowledge.

PROPOSED METHODOLOGY

Evaluation Plan

1 Requirement check

Objective 1

Run the same cases on the Gem tutors and on Socratic AI, and mark R1 to R9.

2 Assessor vs human markers

Objective 2

Mark 61 answers by hand and compare them using Cohen's kappa (target 0.8).

3 With and without agents

All objectives

Switch the Planner, simulations or Checker off, and compare the same sessions.

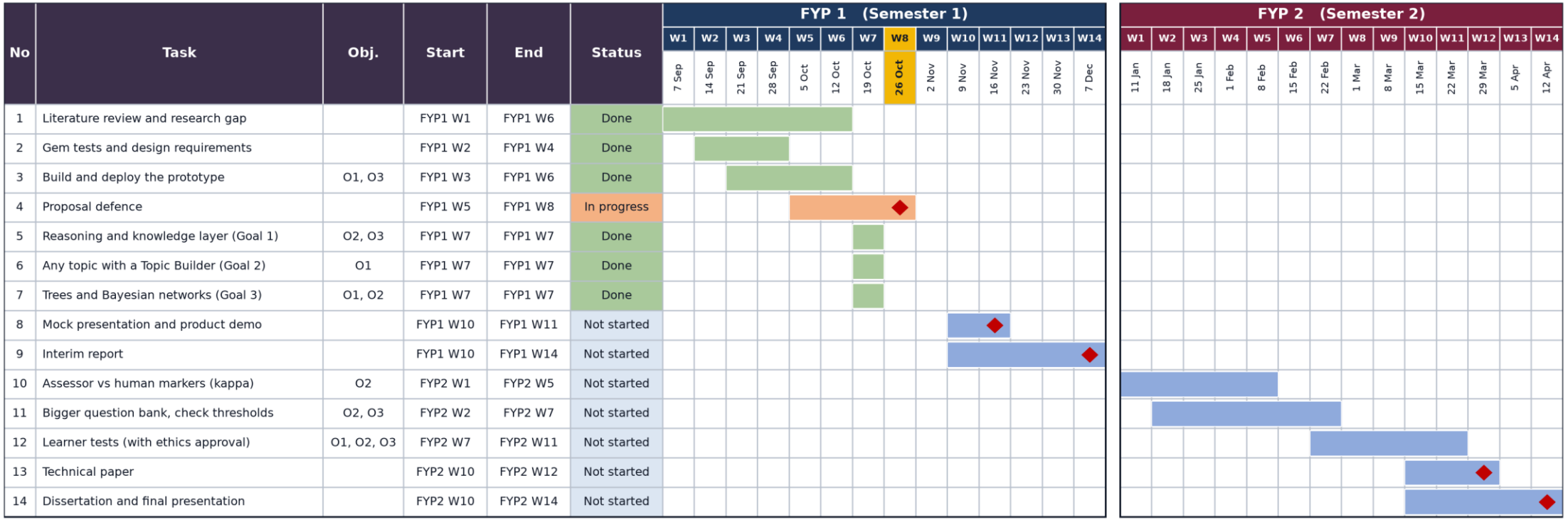
4 Learner experience

Objective 3

A short questionnaire with Dr Ho's students, after ethics approval.

GANTT CHART & MILESTONES

No	Task	Obj.	Start	End	Status
1	Literature review and research gap		FYP1 W1	FYP1 W6	Done
2	Gem tests and design requirements		FYP1 W2	FYP1 W4	Done
3	Build and deploy the prototype	O1, O3	FYP1 W3	FYP1 W6	Done
4	Proposal defence		FYP1 W5	FYP1 W8	In progress
5	Reasoning and knowledge layer (Goal 1)	O2, O3	FYP1 W7	FYP1 W7	Done
6	Any topic with a Topic Builder (Goal 2)	O1	FYP1 W7	FYP1 W7	Done
7	Trees and Bayesian networks (Goal 3)	O1, O2	FYP1 W7	FYP1 W7	Done
8	Mock presentation and product demo		FYP1 W10	FYP1 W11	Not started
9	Interim report		FYP1 W10	FYP1 W14	Not started
10	Assessor vs human markers (kappa)	O2	FYP2 W1	FYP2 W5	Not started
11	Bigger question bank, check thresholds	O2, O3	FYP2 W2	FYP2 W7	Not started
12	Learner tests (with ethics approval)	O1, O2, O3	FYP2 W7	FYP2 W11	Not started
13	Technical paper		FYP2 W10	FYP2 W12	Not started
14	Dissertation and final presentation		FYP2 W10	FYP2 W14	Not started



28 Oct 2026 (FYP1 W8)
Proposal defence

16 Nov 2026 (FYP1 W11)
Product demo, mock presentation

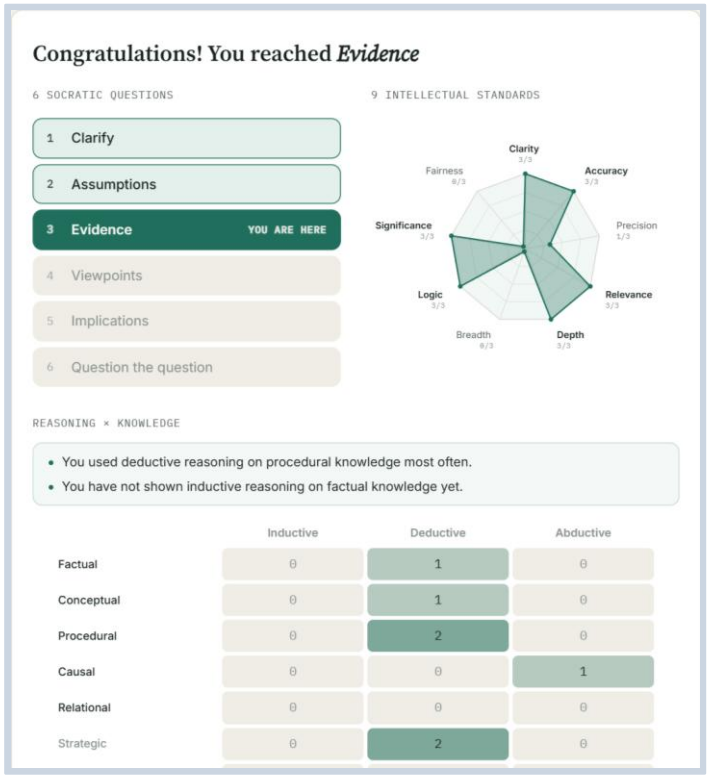
7 Dec 2026 (FYP1 W14)
Interim report

29 Mar 2027 (FYP2 W12)
Technical paper

12 Apr 2027 (FYP2 W14)
Dissertation, final presentation

PRELIMINARY RESULTS

A Real 3-Round Session (Gemini)



The end profile shows the level, the 9 standards and the reasoning and knowledge grid

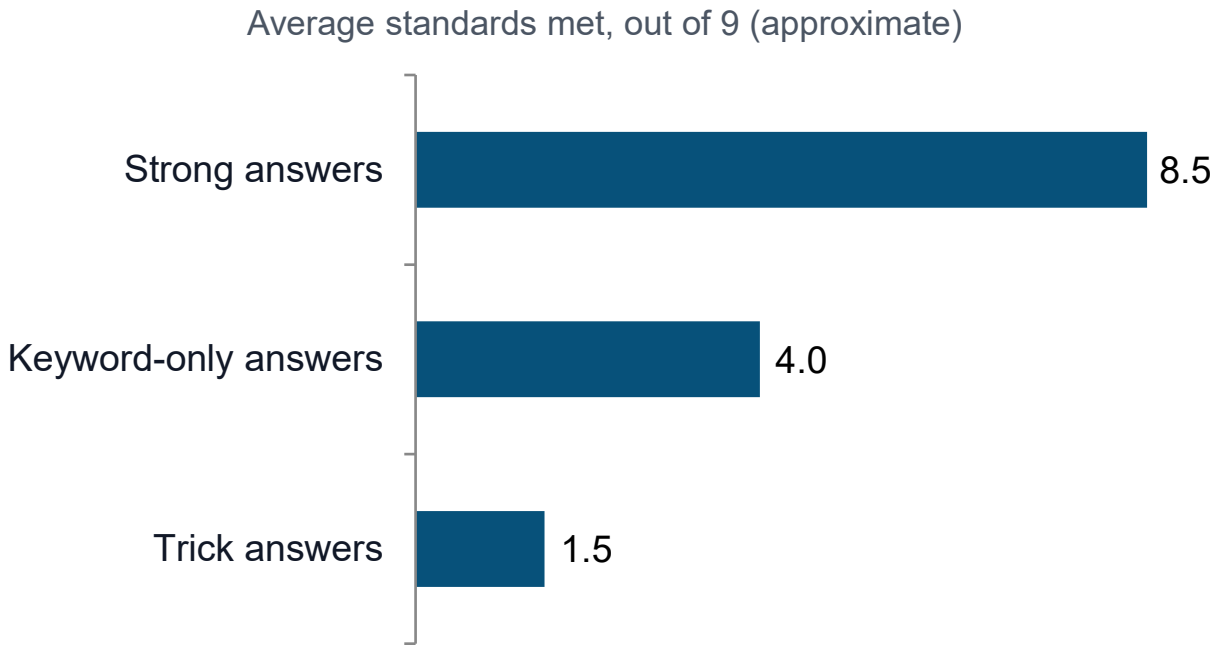
0 times the answer was given away, even when I asked for it

3 of 3 main problems found in testing were fixed

~24 AI calls per session, inside the free tier

PRELIMINARY RESULTS

Assessor on a 61-Answer Test Set



37 real and 24 made-up answers, not yet checked against human markers.

Main problems found and fixed

- The tutor asked the same question in new words, so the Checker now compares questions.
- Levels rose too fast, so a level up now needs 2 good answers.
- The Assessor was too generous, so each standard now has strict rules.

Next, keyword-only answers still score about 4 of 9, so I will make the Assessor stricter.

CONCLUSIONS

The Socratic AI tutors I tested have no **clear question order**, no **measurement** and no **adaptation**.

Socratic AI fixes all three. It asks up Paul's six levels, marks each answer on nine standards and adapts to each student.

It also shows the **type of reasoning** and **knowledge used** on any topic. Next, human markers will check its marks.

Take-home message

Socratic AI trains thinking, it is not an answer machine. AI agents work together, and simple code rules keep them in check.

REFERENCES

[1] R. Paul and L. Elder, The Thinker's Guide to the Art of Socratic Questioning. Foundation for Critical Thinking, 2006; and The Miniature Guide to Critical Thinking Concepts and Tools, 8th ed. Rowman & Littlefield, 2019.

[2] L. W. Anderson and D. R. Krathwohl, Eds., A Taxonomy for Learning, Teaching, and Assessing. Longman, 2001.

[3] J. B. Biggs and K. F. Collis, Evaluating the Quality of Learning: The SOLO Taxonomy. Academic Press, 1982.

[4] H. Sahamid, "Developing critical thinking through Socratic questioning," Int. J. Educ. Literacy Stud., vol. 4, no. 3, pp. 62-72, 2016.

[5] A. Makhene, "The use of Socratic inquiry to facilitate critical thinking in nursing education," Health SA Gesondheid, vol. 24, a1224, 2019.

[6] Y.-R. Ho, B.-Y. Chen and C.-M. Li, "Thinking more wisely: Using the Socratic method to develop critical thinking skills amongst healthcare students," BMC Med. Educ., vol. 23, 173, 2023.

[7] P.-B. Degen and I. Asanov, "Beyond automation: Socratic AI, epistemic agency, and ... multi-agent learning architectures," arXiv:2508.05116, 2025.

[8] F. Fan, "Socratic scaffolding in AI education: A framework for critical thinking," Regional Educ. Technol. J., vol. 8, no. 1, 2025.

[9] K. W. Lee, Y. S. Ang and J. W. Lai, "Learning analytics with scalable Bloom's taxonomy labeling of Socratic chatbot dialogues," Computers, vol. 14, no. 12, 555, 2025.

[10] H.-P. Lee et al., "The impact of generative AI on critical thinking," in Proc. CHI '25, Art. 1121, 2025.

[11] A. Sarkar, "How to stop AI from killing your critical thinking," TEDAI Vienna, 2025.

[12] L. Brand, "A practical guide to agentic AI in higher education," QS, 11 Sep. 2026.

REFERENCES

[13] R. W. Hamming, "Error detecting and error correcting codes," Bell Syst. Tech. J., vol. 29, no. 2, pp. 147-160, 1950.

[14] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," NeurIPS, vol. 33, pp. 9459-9474, 2020.

[15] T. W. Ho, Foundations of Modern Deep Neural Networks, Module 8: Towards Agentic AI [Course slides]. CAPE, Universiti Teknologi PETRONAS, 2025.

[16] J. Wei et al., "Chain-of-thought prompting elicits reasoning in large language models," NeurIPS, vol. 35, pp. 24824-24837, 2022.

[17] S. Yao et al., "ReAct: Synergizing reasoning and acting in language models," in Proc. ICLR, 2023.

[18] Y. Gao et al., "Retrieval-augmented generation for large language models: A survey," arXiv:2312.10997, 2023.

[19] Anthropic, "Introducing the Model Context Protocol," 25 Nov. 2024. [Online]. Available: https://www.anthropic.com/news/model-context-protocol

[20] S. Hong et al., "MetaGPT: Meta programming for a multi-agent collaborative framework," in Proc. ICLR, 2024.

[21] M. H. Nguyen, T. P. Chau, P. X. Nguyen and N. D. Q. Bui, "AgileCoder: Dynamic collaborative agents for software development based on agile methodology," arXiv:2406.11912, 2024.

[22] T. W. Ho, Foundations of Modern Deep Neural Networks, Module 7: Foundation Models and Generative AI [Course slides]. CAPE, Universiti Teknologi PETRONAS, 2025.

[23] DeepLearning.AI, short courses: Knowledge Graphs for RAG; Building and Evaluating Advanced RAG; Agentic Knowledge Graph Construction; Long-Term Agentic Memory with LangGraph; Building Agentic RAG with LlamaIndex; Retrieval Augmented Generation. [Online]. Available: https://learn.deeplearning.ai

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Q & A